

Standard Form & Indices

Test · 41 marks total · Sections F & G include Higher-tier content

Name: _____

Date: _____

Score: _____ / 41

Section A — Laws of indices

A1. Write $4^5 \times 4^3$ as a single power of 4.

[1 mark]

A2. Write $7^9 \div 7^2$ as a single power of 7.

[1 mark]

A3. Write $(6^2)^3$ as a single power of 6.

[1 mark]

A4. Work out the value of $2^3 \times 2^2$.

[2 marks]

Section B — Zero and negative indices

B1. Work out the value of 12^0 .

[1 mark]

B2. Work out the value of 5^{-2} . Give your answer as a fraction.

[2 marks]

B3. Work out the value of 10^{-3} . Give your answer as a decimal.

[2 marks]

Section C — Writing standard form

C1. Write 72,000 in standard form.

[2 marks]

C2. Write 8,300,000 in standard form.

[2 marks]

C3. Write 0.0048 in standard form.

[2 marks]

C4. Write 0.00003 in standard form.

[2 marks]

Section D — Converting back, and calculating

D1. Write 6.5×10^4 as an ordinary number.

[1 mark]

D2. Write 3×10^{-5} as an ordinary number.

[1 mark]

D3. Work out $(4 \times 10^3) \times (2 \times 10^5)$. Give your answer in standard form.

[3 marks]

D4. Work out $(9 \times 10^8) \div (3 \times 10^3)$. Give your answer in standard form.

[3 marks]

Section E — Powers to know, and estimating roots

E1. Write down the value of 3^4 .

[1 mark]

E2. $\sqrt{60}$ lies between which two consecutive whole numbers?

[1 mark]

Section F — Laws of indices with algebra

F1. Simplify $a^6 \times a^4$.

[1 mark]

F2. Simplify $b^{10} \div b^3$.

[1 mark]

F3. Simplify $(c^3)^4$.

[1 mark]

F4. Simplify $5d^2 \times 2d^6$.

[2 marks]

F5. Simplify $(3e^2)^3$.

[2 marks]

Section G — Fractional indices (Higher)

G1. Work out the value of $49^{1/2}$.

[1 mark]

G2. Work out the value of $27^{1/3}$.

[1 mark]

G3. Work out the value of $16^{3/4}$.

[2 marks]

G4. Work out the value of $9^{-1/2}$.

[2 marks]